



CHAPTER THREE

TECHNIQUE FUNDAMENTALS AND SPOTTING

Proper exercise execution is at the core of safe and effective resistance training. Correct execution of exercises with the addition of competent spotting is a pillar of resistance training's contribution to injury prevention.

Technique Fundamentals

There are several commonalities amongst resistance training exercise techniques. Most free weight and machine exercises involve some sort of handgrip on a bar, dumbbell, or handle, and all exercises proceed from an optimal body or limb position, movement range and speed, and method of breathing. Additionally, some exercises may also warrant the use of a weight belt and certain procedures for lifting a bar off the floor (2).

Handgrips

Resistance training involves two main types of handgrips: (a) the pronated grip, with palms down and knuckles up as you face the bar, also called the overhand grip; and (b) the supinated grip, with palms up and knuckles down as you face the bar, also called the underhand grip (2). A neutral grip is the position where the palms face each other—as in a handshake. Neutral grips often apply to dumbbells and some specially designed angled bars. Two less common grips are the alternated grip, in which one hand is in a pronated grip and the other is in a supinated grip, and the hook grip, which is similar to the pronated grip except that the thumb is positioned under the index and middle fingers (2). The hook grip is typically used for performing exercises that require a stronger grasp on the bar due to the higher resistance (e.g., snatch). Note that the thumb is wrapped around the bar in all of the grips shown; this position is called a closed grip. When the thumb does not wrap around the bar, the grip is open, or considered a false grip. Establishing the proper grip in an exercise involves placing the hands at the correct distance from each other (referred to as the grip width) (2).

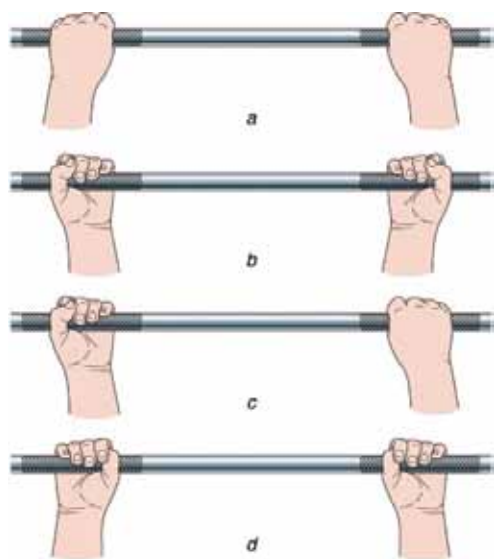


Figure 3-1. Examples of Different Barbell Grips: a) Pronated/Overhand, b) Supinated/Underhand, c) Alternated, and d) Hook

Grip Width

The three main handgrip widths are common, wide, and narrow. For most exercises, the hands are placed at approximately shoulder-width (2). Hand positioning for all exercises should result in a balanced, even bar (Figure 3-2).

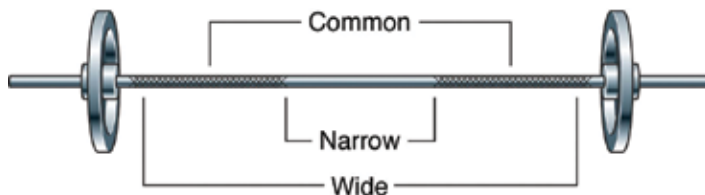


Figure 3-2. Grip Width Landmarks on the Barbell

Stable Body and Limb Positioning

Whether an exercise requires lifting a barbell or dumbbell from the floor, or pushing and pulling while on a machine, establishing a stable position is critical. A stable position enables the athlete to maintain proper body alignment during an exercise, which in turn places the appropriate stress on the target muscles. Exercises performed while standing typically require that foot position be slightly wider than hip-width with the heels and balls of the feet in contact with the floor. Establishing a stable position in, or on, machines sometimes requires adjusting the seat or movable resistance arm, fastening belts snugly, or correctly placing movable pads. Seated, or supine, exercises performed on a bench require a specific posture. The athlete should position the body to achieve a five-point body contact position:

1. Head in neutral position on bench or back pad
2. Shoulders and upper back set evenly on bench or back pad
3. Buttocks on bench or seat
4. Right foot flat on floor
5. Left foot flat on floor

Cam-, pulley-, or lever-based exercise machines that have an axis of rotation require specific positioning of the resistance and the athlete's arms and/or legs for reasons of safety and optimal execution posture. Align the target joint, and thus its muscle group(s) involved in the exercise, with the axis of the machine. Proper machine placement may require seat, roller pad, thigh pad, back pad, or chest pad adjustment.

Range of Motion and Speed

Nearly all exercises use the full range of motion of a joint or machine. Advanced training approaches (not covered here) sometimes use less than the full range of motion. When the goal is power or explosive exercise performance, the effort should accelerate the resistance to a maximal, controllable speed. Some machines do not permit rapid movements. Always check the manufacturer's instructions to determine a machine's proper use.

Breathing Considerations

The most strenuous position in the range of motion of an exercise is referred to as the “sticking point” (typically soon after the transition from the eccentric phase to the concentric phase) (2,3). Instruct athletes to exhale through the sticking point and to inhale during the less stressful phase of the repetition. Breath holding (called the Valsalva maneuver) is appropriate in some exercise situations. The Valsalva maneuver is reserved for advanced athletes, and can be helpful for maintaining proper vertebral alignment and support (1,2). The Valsalva maneuver involves expiring against a closed glottis, which, when combined with contracting the abdomen and rib muscles, creates more rigidity in the abdominal area to help stabilize the spine (1,2). The resulting increase in intra-abdominal pressure has potentially detrimental side effects, such as dizziness, disorientation, excessively high blood pressure, and blackouts. Breath holding should be transient, about 1 – 2 s, and allow the athlete to stabilize the spine during the high exertion range of motion and then breathe again.

Lifting a Bar from the Floor

Proper position of the feet and back enables the leg muscles to provide the major contribution for lifting from the floor. It is important to keep the bar close to the body and the back flat during the upward movement. Encourage the athlete to bend at the knees more than at the hips in order to lift the resistance from the floor.

Spotting

A spotter is someone who assists in the execution of an exercise to help protect the athlete from injury. Spotting resistance exercises is a skill. As a skill, spotting requires instruction and practice. A spotter may also serve to motivate the athlete and help in partner-assisted repetitions. However, the spotter’s primary responsibility is the safety of the athlete.

Types of Exercises that Require Spotting

Free weight exercises performed over the head (e.g., barbell shoulder press), with the bar on the back (e.g., back squat), with the bar racked anteriorly on the front of the shoulders or on the clavicles (e.g., front squat), and over the face (e.g., bench press, lying triceps extension) are challenging for the athlete. When lifting heavy resistances, the role of the spotter increases due to the potential inability of the athlete to lift the weight to a safe position. Some specific exercises or lifts may require more than one spotter. For example, weights lifted to the side, or too heavy resistances for a single spotter to handle, may require the aid of other spotters to watch the weight(s) while another spotter assists the athlete. Vigilance is required whenever the bar is over the head, over the face, or otherwise unstable. Spotting dumbbell exercises typically requires more skill than spotting barbell exercises because there are two pieces of equipment to control. Contraindications for spotting include power exercises at high speeds.

Spotting Overhead Exercises

Ideally, to promote safety of the athlete during lifts with the bar on the back, or front of the shoulders, a power rack with the safety crossbars in place at an appropriate height is recommended. Clear all objects that might cause tripping from the lifting area. One of the most common causes of injury in a weight training area is tripping over a weight plate on the floor. Nonparticipating athletes should stay clear of the lifting area and avoid distracting the lifter’s attention by standing or moving in front of the athlete. Spotters must be strong enough and tall enough to catch unstable weights and match the height of the athlete.

Spotting Over-the-Face Exercises

The spotter should grasp the bar with an alternated grip, usually narrower than the athlete’s grip when spotting an over-the-face exercise. The spotter should use an alternated grip to pick up the bar and return it to the floor, or rack, and a supinated grip to spot the bar during the lift. The grips described help ensure that the bar does not roll out of the spotter’s hands and onto the athlete’s face or neck. For dumbbell exercises, the spotter should spot as close to the dumbbells as possible. In some dumbbell exercises, it is more practical to spot the arms of the athlete rather than the dumbbells. Spotting at the forearms near the wrists provides a safer technique in these situations. Note that for some exercises (e.g., dumbbell pullover and overhead dumbbell triceps extension) it is necessary to spot with hands directly on the dumbbells.

Spotting Considerations for Power Exercises

Spotting recommendations change for power exercises. Instead of spotting power exercises, the coach should teach the athlete how to get away from a bar that is out of control or unstable. Spotting these types of exercises is too dangerous to both the spotter and athlete. When the bar is in front of the athlete, the athlete should miss the lift by pushing the bar away or simply dropping it. If the bar becomes unstable behind the head, the athlete should release the bar and jump forward. For these reasons, a clear area and platform are necessary to avoid injury to the athlete and others.

Number of Spotters

The number of spotters is determined directly by the specific exercise and the load. The experience and ability of the athlete and spotters, strength of the spotters, and mutual experience also helps determine the number of spotters. Obviously, with heavier loads, the likelihood of injuries increases. Once the load exceeds a single spotter’s ability to effectively protect the athlete (and him- or herself), incorporate another spotter.

Communication between Athlete and Spotter

Communication is a shared responsibility. Before beginning a set, the spotter(s) and athlete should establish and communicate how initial bar movements will occur, the intended number of repetitions, and signals for moving the bar into position. If the spotters do not have this information, they may take control of the bar improperly, too soon, or too late, and consequently disrupt the exercise or injure the athlete or themselves.

Amount and Timing of Spotting Assistance

Sometimes less is more. In addition to protecting the athlete from a fall or dropped weight, a spotter should know how much to assist the athlete when the situation demands only marginal assistance. At the first indication that a repetition is failing, the athlete should quickly ask or signal the spotter (sometimes with just a grunt or sound) for help, and the spotter needs to provide the amount of assistance needed for safe completion. If the athlete cannot contribute anything to the completion of the repetition, the athlete should immediately tell the spotter to “take it” or use a similar phrase. The spotter should take the bar from the athlete quickly and smoothly.

Spotting Techniques

Note that the exercises in this section serve as a model for spotting technique. Exercise technique for each exercise will be addressed in the next chapter.

Barbell Bench Press – Spotting Technique

Stand erect and very close to the head of the bench (but do not distract the athlete)

- Place feet shoulder-width apart with knees slightly flexed
- Grasp the bar with a closed, alternated grip inside the athlete’s grip (Figure 3-3)
- Assist with moving the bar off the supports at athlete’s signal (Figure 3-3)
- Guide the bar to a position over the athlete’s chest and release the bar smoothly (Figure 3-4)
- Full attention should be given to the athlete through all repetitions of the set (Figure 3-5)
- At the end of the set, the athlete will signal for assistance in racking the bar (Figure 3-6)
- Keep a grip on the bar until it is racked (Figure 3-6)



Figure 3-3. Assisted Lift-Off



Figure 3-4. Lifting Position



Figure 3-5. Lowest Lifting Position



Figure 3-6. Racking Position

Dumbbell Incline Bench Press – Spotting Technique

- Stand erect and very close to the head of the bench (but do not distract the athlete)
- Place feet shoulder-width apart with knees slightly flexed
- Place hands near the athlete's wrists (Figure 3-7)
- Keep hands near the athlete's wrists and follow the path of the lift as the dumbbells descend and ascend (Figure 3-8)
- Grasp the athlete's wrists if they begin to struggle and/or get outside the path of the lift (Figure 3-9)



Figure 3-7. Start Position



Figure 3-8. Bottom Lifting Position



Figure 3-9. Spotter Assisting Ascent



Figure 3-10. Start Position



Figure 3-11. Highest Position



Figure 3-12. Spotter Assisting Descent

Barbell Standing Behind the Neck Shoulder Press – Spotting Technique

- Stand directly behind the athlete and place hands outside of the athlete's grip (Figure 3-10)
- Keep hands close to the bar, but not touching it, as the bar ascends and descends (Figure 3-11)
- If the bar descends too fast, or the athlete becomes off-balanced, assist in keeping the descent and movement slow and controlled (Figure 3-12)

Barbell Back Squat – Spotting Technique with One Spotter

- Stand behind the athlete at a point where they will back out from the rack (Figure 3-13)
- Step up to the athlete after they have set their stance, set feet slightly wider than theirs, and position arms under the athlete's armpits, next to their side (Figure 3-14)
- As the athlete descends, descend in unison keeping arms under their armpits without touching (Figure 3-15)
- As the athlete ascends, ascend in unison keeping arms under their armpits without touching (Figure 3-16)
- Be ready to grasp the athlete around the torso if they fail to maintain proper technique and/or complete a repetition
- If the athlete fails, wrap arms under their armpits and around their torso to help them return to an erect position (Figure 3-17)
- If the spotter assists in a repetition, the athlete and spotter should walk the weight back to be racked (Figure 3-18)



Figure 3-13. Athlete Lift-Off



Figure 3-14. Start Position



Figure 3-15. Lowest Squat Position



Figure 3-16. Ascending



Figure 3-17. Spotter Assisting Ascent



Figure 3-18. Racking the Bar

Barbell Back Squat – Spotting Technique with Three Spotters

- The main spotter will stand behind the athlete at a point where they will back out from the rack
- The two side spotters will stand on each end of the barbell
- The main spotter will step up to the athlete after they have set their stance, set feet slightly wider than theirs and position arms under the athlete's armpits, next to their side
- The two side spotters will be in a "ready position" with their hands cupped near each end of the barbell (Figure 3-19)
- As the athlete descends, descend in unison keeping arms under their armpits without touching them
- Side spotters will descend evenly with the athlete keeping their hands near the ends of the barbell
- As the athlete ascends, ascend in unison keeping the arms under their armpits without touching them
- Side spotters will ascend evenly with the athlete keeping their hands near the ends of the barbell (Figure 3-20)
- The main spotter should be ready to grasp the athlete around the torso if they fail to maintain proper technique and/or complete the repetition, and side spotters should be ready to grasp the ends of the barbell
- If the athlete fails, the main spotter will yell "take it" and wrap their arms around the torso to help the athlete return to an erect position
- The side spotters will simultaneously grab each end of the barbell to assist the main spotter and athlete if a repetition is failed. All three spotters should assist evenly to ensure the safest condition for the athlete and spotters
- If the spotters assist in the repetition, the athlete and all three spotters should walk the weight back to be racked



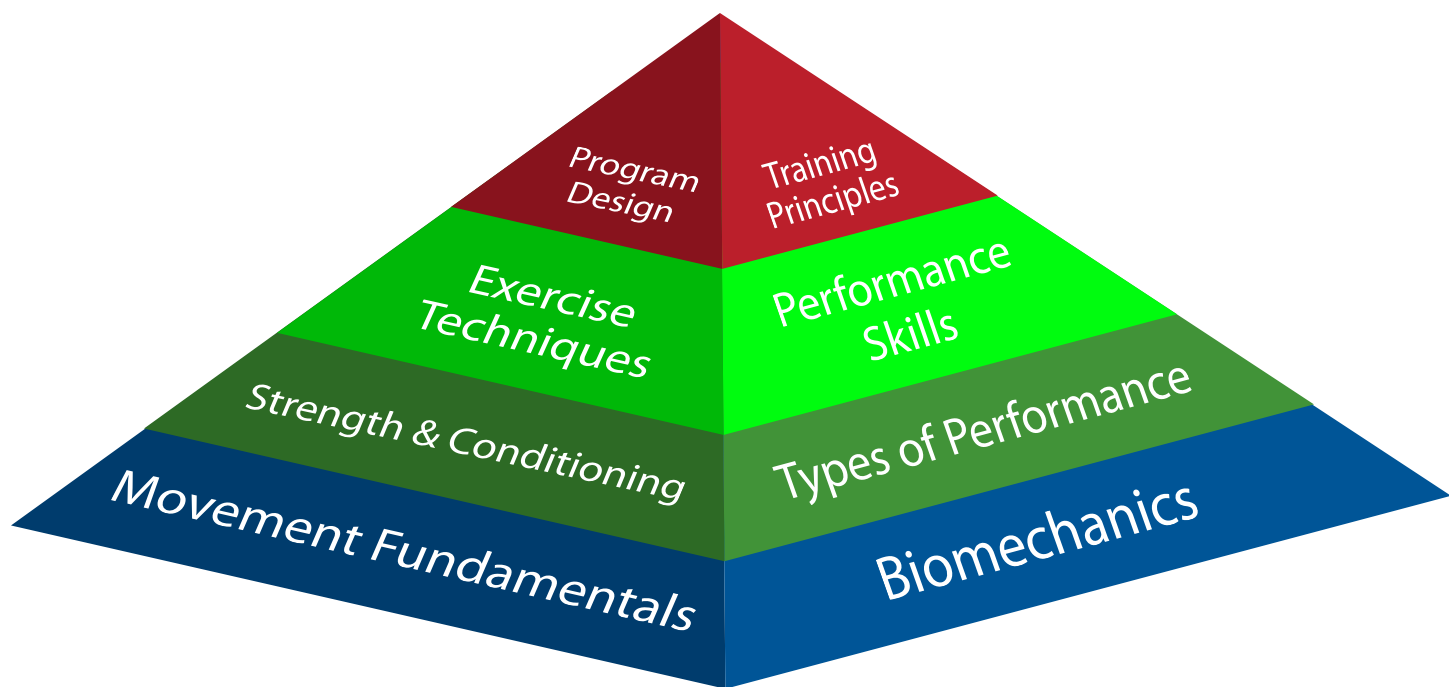
Figure 3-19. Start Position



Figure 3-20. Lowest Position

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